

PS3 cluster creates homemade, cheaper supercomputer

UMass professor joins eight PlayStation cell chips to run highly complex calculations

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A professor at the University of Massachusetts wasn't just goofing off when he started hooking up one PlayStation 3 (PS3) system after another.

Gaurav [Khanna](#), an astrophysicist working at the university's Dartmouth campus, has been building his own supercomputer of sorts with the popular gaming console.

The professor has been renting time on supercomputers at [NASA](#) and the [National Science Foundation](#) to run highly complicated calculations on the amount of radiation emitted when a black hole swallows a star. That supercomputing time, though, doesn't come easily or cheap, Khanna said. In an average year, he rents about 30,000 hours, which costs between \$20,000 and \$30,000, a significant chunk of his grant money.

To ease his supercomputing plight, Khanna turned to the cell chip inside the PS3. By linking eight of them together, he said he gets the same processing power as a supercomputer with 200 processors.

"For \$4,000 or so, I can get eight PS3s that can do the same task that I'd do on a supercomputer," he said. "For a one-time cost, I have this resource I can use privately. I can use it indefinitely over and over again. That's hugely attractive. That's why I considered the project. I have my own supercomputer right here. There's no elaborate process for getting time. There's no waiting. It's just mine."

Khanna said he spent a month testing the system to make sure it was giving him accurate calculations, and he has been using it in actual research for the past two weeks.

Calculations that would take nearly a year to finish on a desktop computer are being done by the cell chip array in less than a day.

Khanna said the cell chip, which was collaboratively developed by Sony Corp., IBM and [Toshiba Corp.](#), is designed to have a single processor and eight compute engines, making it more powerful than dual-core or even quad-core chips. The engines don't handle input/output. They don't talk to the keyboard, mouse or memory. They do raw computations very, very fast, according to Khanna.

"It has a unique design. It has a lot more potential -- a better processor in my opinion," he added.

"What's unique is that they made it an open platform. Normally with a game console, the maker controls who can run what on it. What Sony did was make the PS3 an open platform. They let you run whatever you want on it. It has the full capabilities of a normal computer. You can run [Firefox](#) or

whatever you want. It gave me the possibility of doing whatever I want with it."

Dan Olds, an analyst at the Gabriel Consulting Group, said this is the type of technology that IT managers should be keeping an eye on.

"It sounds like a real wild thing and people will wonder why the hell it works," said Olds. "It works because that particular video game processor is a supercomputer processor. It will scream on the right kinds of workloads. The things that make it really good for video game processing make it good on certain processing-intensive workloads. This would be good for doing highly parallel processor-intensive analysis, taking on a big bunch of numbers, big calculations. It would not be good at running databases because the systems don't have the kind of input/output necessary."

Olds pointed out that IBM already picked up on this capability and integrated the cell chip into its QS20 BladeCenter product. An update to that -- the QS21 -- is slated to ship on Friday. IBM noted that the QS21 is focused on companies that create and run highly visual, graphic and immersive, real-time applications like 3-D rendering, compression and encryption.

"We're starting to move into an era where your average company can get supercomputing power," said Olds.